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Experimental heat transfer study of confined flame jet impinging on a flat surface

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ABSTRACT

An experimental heat transfer study has been conducted for a confined turbulent impinging flame jet. The flame jet was subsonic or supersonic depending on the operating conditions. The impinged target was the flat surface of a water-cooled calorimeter, custom designed to sustain high heat fluxes and high surface temperatures. In order to influence the surface temperature of the target and to observe for the first time its effect on the heat transfer when impinged by the same flame jet, two calorimeters, made of different materials, were applied. Additionally, the heat transfer study presents the influence of process parameters on recorded heat flux for firing rates between 23 and 54 kW, dimensionless nozzle-to-target distances between 5 and 14 and exit nozzle diameters of 7.9 and 12.1 mm. The maximum recorded surface temperature was 600 °C and the maximum recorded heat flux was 10.5 $\frac{MW}{m^2}$. The obtained results were in agreement with heat transfer studies of unconfined impinging flame jets presented in literature. The results showed that if the target surface temperature was increased from around 170 °C to around 530 °C it did not influence the heat transfer coefficient considerably. Also, it was concluded that the influence of the confinement was mostly reflected in preventing the underexpansion of supersonic flame jet.

1. Introduction

Heat stored in the Earth can be exploited through a full-scale geothermal well that has to be drilled deep within the ground through hot, hard and abrasive crystalline rock formations [1,2]. Conventional drilling technologies are very time and cost consuming under these circumstances [2], thus, new ones are under development. Flame-jet drilling (also called thermal spallation drilling or jet piercing) is one of the alternative technologies with potential for drilling in an energy-efficient and cost-effective way [3]. It is based on inducing a steep temperature gradient in the surface rock layers by an impinging flame jet. The temperature gradient causes thermal stresses, localized fracturing and detachment of small chips (spalls) from the rock surface [4,5]. Continuous fracturing and spalls removal consequently results into a cavity.

The flame jet drilling technology can be evaluated, understood and improved not only by analyzing the drilled rock, but also by describing the impingement heat transfer between the flame jet and the rock surface. The heat transfer study is necessary to define the requirements for the impinging flame jet in the drilling process and to map the boundary conditions for thermal stress analysis in the rock sample. Therefore, the influence of process parameters (e.g. firing rate, nozzle diameter, nozzle-to-target distance) and target surface temperature on the heat transfer has to be investigated. The flame jet had to be confined in order to simulate the physical conditions existing in a borehole. Also, in the required heat transfer study the surface temperatures of the target have to be similar to the temperatures of the rock surface at the onset of spallation (between 200 and 650 °C) [6].

Development of flame jet drilling in the past and its optimization was performed by Browning et al. [7]. According to the experimental results the released heat from the flame has an impact on spall thickness and volumetric removal rate. A reasons is that the heat transferred through the rock surface influences the development of thermal stress and finally the drilling itself. Only the heat released through the nozzle was reported. The heat transfer to a target, which improves the understanding of the phenomenon, was not characterized in detail.

Heat transfer from an impinging flame jet applied in flame jet drilling experiments was further studied by Rauenzahn [8]. He performed a heat transfer study with a small quarry slotting tool producing supersonic flame jets. The purpose of the performed heat transfer study was to measure the boundary conditions required for a flame jet drilling model. The heat flux was calculated according to an energy balance of a

Abbreviations: *AC*, alternating current; *FJD*, flame jet drilling

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Nomenclature

Roman

A	area (m^2)
b_0,b_1	regression coefficient
D,D_i	nozzle diameter (mm)
C_p	heat capacity (J/kg K)
h	heat transfer coefficient (W/m^2 K)
L	dimensionless distance (–)
$\dot{m}$	mass flow (kg/s)
p_{ch}	chamber pressure (bar)
$\dot{q}$	heat flux (W/m^2)
$\dot{q}_{real}$	heat flux through the sensing surface (W/m^2)
$\dot{q}_{sen}$	heat flux that entered the calorimetric cooling water (W/m^2)
$\dot{Q}_f$	firing rate (kW)
S	distance between exit nozzle and target (m)
T	temperature (°C)
U	various variables

Greek

λ	air-fuel equivalence ratio (–)
σ	standard deviation
δ	uncertainty

Subscripts

d	device
f	flame
guard	guard
inlet	inlet
max	maximum
nozzle	exit nozzle
outlet	outlet
surf	surface

copper block impinged by the flam jet. The block had embedded thermocouples and the impinged surface was concave. The tested process parameters, recorded temperatures of the copper block and heat fluxes are presented in Table 1.

As noticed by Rauenzahn, during the performed heat transfer study difficulties occurred while measuring temperatures within the copper block and while calculating the heat transferred to the concave surface. A prediction of the hole diameter was very successful, but the predicted heat transfer rates were not aligned with the experimental data.

Only Rauenzahn [8] studied a confined impinging flame jet and for limited conditions as listed in Table 1. But, there are numerous heat transfer studies dedicated to free flame jets impinging a flat surface. For the free impinging flame jets the heat transfer is dependent on the diameter of a burner exit nozzle, axial distance between the target and the nozzle, the released flame power and adiabatic flame temperature [9,10]. In those studies mostly laminar flame jets reaching Reynolds numbers up to 2500 were observed [11–13]. Van der Meer reached Reynolds numbers up to 4226 [14] when comparing the heat transfer from isothermal and flame jets. Higher Reynolds numbers (up to 35,300) were reported by Milson et al. [15] who observed the heat transfer on a steel plate at constant nozzle-to-target dimensionless distance. As a fuel, mostly methane [16,11] and natural gas [14] were combusted. The largest firing rate reported in literature was applied in the studies from Baukal et al. [16] reaching 25 kW. They concluded that increasing the firing rate by 400% the heat flux to the target increases by 470%. Concerning the influence of the axial distance on the heat transfer the same trend was noticed by many researchers: the maximum heat transfer occurs at the end of the jet potential core [11,13–15]. The only work investigating the influence of the nozzle diameter was presented by Chander et al. [11], who performed the study for the laminar regime. They concluded that the heat transfer was affected by the nozzle diameter only in the wall jet region.

The target in most heat transfer studies is a heat flux sensor. It is designed usually depending on the flame properties (temperature, exhaust gas velocity, firing rate) in order to sustain the harsh environment. Examples include a water-cooled calorimeter [16,17], a Gardon heat flux transducer [14] and a Vatell heat flux sensor [11]. The highest heat flux recorded in heat transfer studies of impinging free flame jets was 1 $\frac{MW}{m^2}$ [16].

Both, the heat flux measuring device and the heat transfer study of the confined impinging flame jet are presented in this work. The targets for the heat transfer study were two custom designed water-cooled calorimeters that recorded surface temperatures between 70 and 600 °C. According to Baukal et al. [9] no flame jet impingement heat transfer study with target surface temperatures between 350 and 700 °C has been published yet. The influence of different surface temperature on the heat transfer from an impinging flame jet has also never been studied before and is presented in this work. Process conditions that successfully drilled rock samples produce subsonic and underexpanded supersonic flame jets, thus, both cases were studied. Heat fluxes up to 10.5 $\frac{MW}{m^2}$ were measured, which is higher than any reported heat fluxes in literature. The heat flux transferred to the target was adjusted by changing the firing rate, the nozzle diameter and axial dimensionless distance. The influence of confinement on the heat flux transferred from the impinging flame jet is discussed.

2. Experimental apparatus and measurement technique

The flame jet drilling (FJD) setup applied in this heat transfer study consisted of two main parts: a burner and a heat flux sensor. The flame jet was exiting the burner and it was impinging the top of the sensor. The sensor was cooled with water flowing beneath its surface and absorbing the transferred heat. A schematic overview of the experimental system is presented in the Fig. 1. The whole setup was placed in an insulation box that was ventilated with 5000 $\frac{m^3}{h}$ of air.

2.1. Burner

In the burner (see Fig. 2) a premixed methane-air gas mixture was combusted. The gases entered the mixing pipe with a perforated 5 mm thick plate installed perpendicular to the gas flow at the top of the burner. After the mixing pipe the gas mixture entered the combustion chamber where ignition took place. It was initiated by a ceramic heater (Type EF, Bach RC, Werneuchen, Germany) made of silicon nitride (Si_3N_4) doped yttrium oxide (Y_2O_3) having a resistance of 40 Ω at 20 °C. It was connected to alternating current (AC) power supply and its

Table 1
Heat transfer study performed by Rauenzahn et al. [8].

Property of a flame jet	Measured value
Distance to target [cm]	7.6,10.2,12.7,15.2
Fuel flow rate $\left[\frac{l}{s}\right]$ STP	1
Oxygen flow rate $\left[\frac{l}{s}\right]$ STP	5.1
Max. target temp. [°C]	266
Max. heat flux $\left[\frac{MW}{m^2}\right]$	7.4

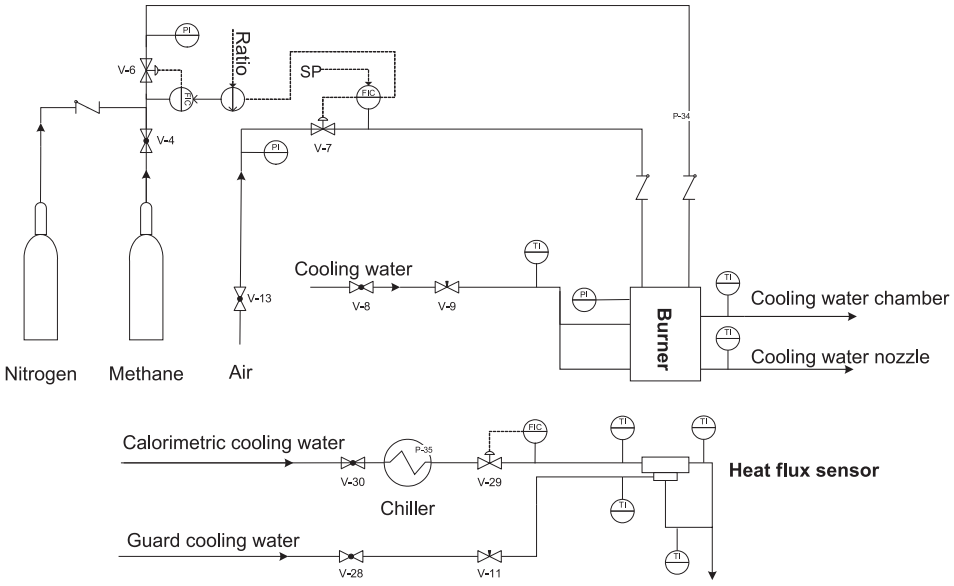


Fig. 1. Schematic overview of the flame impinging system (Flame jet drilling setup and heat flux sensor (calorimeter)). Methane was used as a fuel, air as oxidizer and nitrogen to purge the lines at the end of experiment. The flame jet impinged the calorimeter from above. The burner and the heat flux sensor were cooled with water.

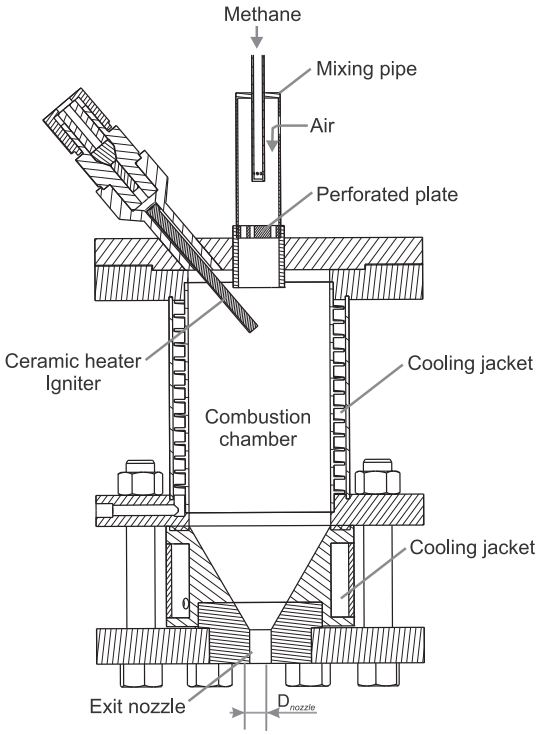


Fig. 2. Schematic overview of the burner. Methane and air were mixed in the mixing pipe. The gas mixture entered the combustion chamber, which was ignited by a ceramic heater. The combustion products were exiting the burner at the bottom through a replaceable exit nozzle.

temperature was adjusted by means of an AC transformer. The electrical power required for ignition of the gas mixture was 205 W. The air-fuel equivalence ratio (λ) was equal to 1.1 in all experiments. The flame power (firing rate $\dot{Q}_f$) was varied between 23 and 54 kW by adjusting the air and methane mass flow rate with two PID controllers. The firing rate, or gross heat release rate, was calculated according to Eq. (1) as suggested by Baukal [18]:

$$\dot{Q}_f = \dot{m}_{CH_4} \cdot HHV \tag{1}$$

where $\dot{m}_{CH_4}$ presents the mass flow rate of methane in $\frac{kg}{s}$ and HHV is the higher heating value of methane (55.5 $\frac{MJ}{kg}$). The wall of the combustion chamber was cooled with water of constant mass flow in order to protect the material from overheating and to extend the life time of the burner. The combustion products were directed to the exit nozzle at the bottom of the burner. Two nozzles were tested having a cylindrical shape with a diameter (D) of 7.9 mm (D_1) and 12.1 mm (D_2). Depending on the nozzle diameter and the firing rate the pressure in the combustion chamber (p_{ch}) varied between 1.54 and 3.25 bar. The exit gas velocity was approximated assuming that the fuel was completely combusted, that the flow was steady, one dimensional, adiabatic and that the exhaust gases are ideal. The calculated values were larger than 500 $\frac{m}{s}$ and the corresponding Reynolds numbers larger than 16,000 indicating that the flow was in turbulent regime. The properties of the exhaust gas components were taken from NIST Standard Reference Database. The temperature at the nozzle (T_{nozzle}) was measured with tips of the thermocouples placed at the center of the nozzle exit. There were applied two measuring methods already described in literature by Baukal [19]. The first method was performed with a bare S-type thermocouple (diameter of soldered spot was 1.6 mm) which can sustain present temperatures and the obtained values were corrected as suggested by Siewert [20]. The correction was necessary because of the heat losses due to radiation, convection and conduction of the thermocouple. To reduce the radiation and catalysis in eventual combustion reaction at that location, the thermocouple was coated with Ceramabond 569 as described by Burton et al. [21]. The second method was performed by means of K-type thermocouples (thickness of 1,2 and 3 mm) where extrapolation has been performed for an hypothetically zero diameter thermocouple, thus no temperature correction was necessary. Both methods gave temperatures differing up to around 100 °C, therefore the mean temperature of the results obtained with both methods was used as flame temperature in this study. The operating conditions of the experiments are summarized in Table 2. The recorded

Table 2
Process parameters in the heat transfer studies.

$\dot{Q}_f$ [kW]	D [mm]	L [–]	p_{ch} [bar]	T_{nozzle} [°C]
23	7.9	5–14	1.54	1269
28	7.9	5–14	2.00	1397
33	7.9	5–14	2.54	1481
38	7.9	5–14	3.25	1593
38	12.1	5–14	1.53	1729
54	12.1	5–14	2.03	1801

temperature at the nozzle (T_{nozzle}) was increasing with the firing rate. The most probable reason is that the burning chamber was cooled all the time with a constant cooling water mass flow and the part of the released heat passing through the exit nozzle could increase. Therefore, in case of smaller firing rate the heat transfer to the cooling water was more efficient and the exit nozzle temperature was more influenced being lower than at higher firing rate. The distance between the exit nozzle and the target was presented with L and it was calculated as in Eq. (2).

2.2. Target

The target assembly was designed as presented in Fig. 3. It consisted of a water-cooled calorimeter and a confining ring. The water-cooled calorimeter was used to measure heat flux and it was placed below the exit nozzle. Part of the jet close to the impinged surface was confined with the removable confining ring that was mounted on the top of the calorimeter. The dimensionless distance (L) was calculated as in Eq. (2):

$$L = \frac{S}{D} \tag{2}$$

where S presents the distance between the nozzle exit and the calorimeter surface.

2.2.1. Measurement principle of a water-cooled calorimeter

A water-cooled calorimeter is a device that measures the quantity of absorbed thermal energy from the energy balance equation of the cooling water. In general the heat flux $\dot{q}$ through the calorimeter is calculated at steady-state condition as presented in Eq. (3) [22]:

$$\dot{q} = \frac{\dot{m}C_p}{A}(T_{outlet} - T_{inlet}) \tag{3}$$

where $\dot{m}$ is the mass flow of the cooling water, C_p the heat capacity of water, A the sensing surface area and T_{inlet},T_{outlet} are inlet and outlet temperatures of the cooling water, respectively.

2.2.2. Design of the water-cooled calorimeter

The calorimeter consisted of two parts: (1) sensing element and (2) guard, which are presented in Fig. 3. The sensing element was the central part of the calorimeter. It was surrounded by the guard that was isolating the sensing element from potential heat loss or gain. Both parts were cooled with water: calorimetric water and guard water, respectively.

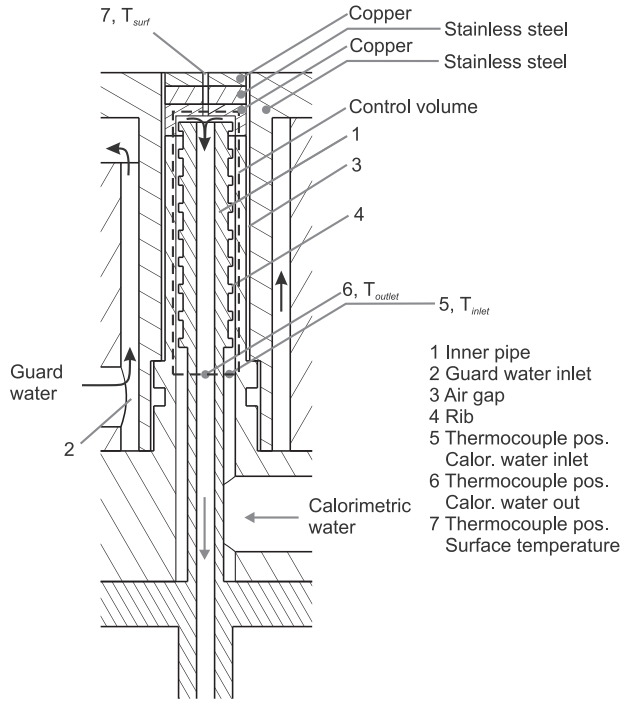


Fig. 4. Closer look at the cross-section of the sensing element. The heat balance equation of the calorimetric cooling water was valid for the marked control volume. The arrows show the flow direction of cooling water (calorimetric and guard water).

In order to simulate conditions which are present in drilling laboratory experiments a removable confining ring was designed (inner diameter 40 mm, height 35 mm) and placed on the top of the calorimeter (see Fig. 3).

A closer look at the cross-section of the sensing element is presented in Fig. 4. It had an empty inner space 5 mm beneath the sensing surface. This is the area where the calorimetric water was flowing. A pipe (1, Fig. 4) was placed along the centerline axis in order to define the direction of the cooling water flow. Thus, the cooling water entered the inner space at the bottom left corner and it was flowing upwards along the outer side of the pipe. At the top of the inner space the water was

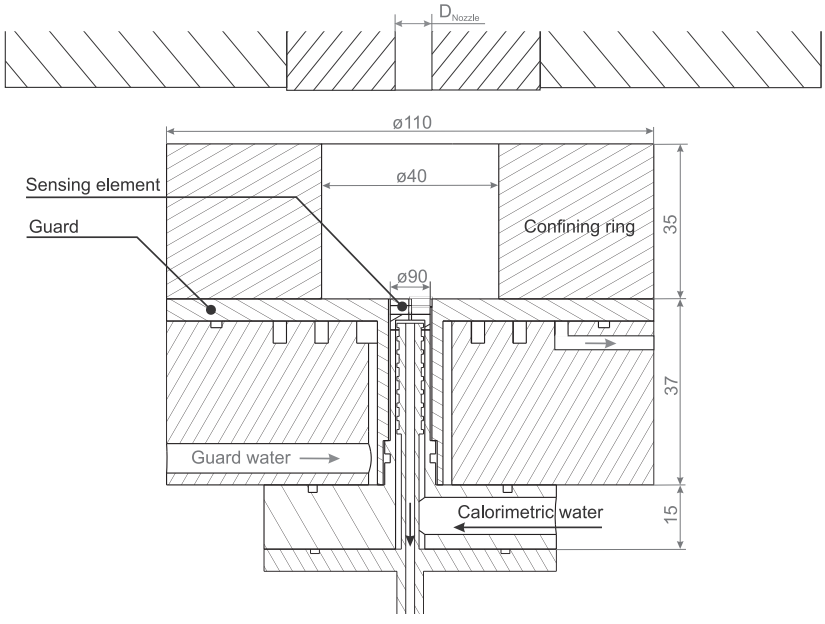


Fig. 3. Design of the heat flux sensor (calorimeter) that was impinged by the flame jet. The measuring method is based on calorimetric measuring principle [22]. It consisted of two parts: sensing element and guard. The confining ring was removable.

entering the pipe flowing to the exit at the bottom through the pipe (6, Fig. 4). The guard water was also flowing 5 mm beneath the guard surface. The water entered the guard inner space at the bottom right corner (2, Fig. 4) flowing upward nearby the sensing element. After reaching the top of the inner space, the guard water was flowing through a spiral channel beneath the surface to the exit located on the side of the guard.

The heat flux measured with the calorimeter was the heat that entered the calorimetric cooling water ($\dot{q}_{sen}$). Ideally, this heat was equal to the heat transferred through the sensing surface ($\dot{q}_{real}$). But, in case of additional heat transfer ($\delta\dot{q}$) (a) from the guard to the sensing element ($\delta\dot{q} > 0$), (b) from the sensing element to the surroundings (e.g. guard, $\delta\dot{q} < 0$),X or c) heat transfer to the cooling water outside of the control volume, $\dot{q}_{sen}$ can be described with Eq. (4):

$$\dot{q}_{sen} = \dot{q}_{real} + \delta\dot{q} \tag{4}$$

In order to prevent heat transfer in radial direction between the sensing element and the guard, a 30 mm long and 0.5 mm thick air gap was introduced (3, Fig. 4). The heat transferred from the sensing element to the calorimetric water was increased with turbulence in the water flow by ribs at the inside pipe (4, Fig. 4). In addition, the wall thickness of the sensing element was minimized, which also increased the transferred heat to the cooling water. The operating flow rate was $2.0\frac{kg}{min}$ and $2.16\frac{kg}{min}$ for the calorimetric and guard water, respectively. The inlet temperature of the calorimetric water was reduced with a chiller before entering the sensing element in order to prevent eventual boiling.

The inlet temperature of the calorimetric water (T_{inlet}) was obtained with three K-type thermocouples (0.5 mm diameter thickness). They were positioned shortly after the water was entering the sensing element around the pipe in the inner space (5, Fig. 4). T_{inlet} from Eq. (3) presents the average of the recorded values of all three thermocouples. The outlet temperature of the calorimetric water (T_{outlet}) was measured with one thermocouple placed in the pipe (6, Fig. 4). The tips of all thermocouples were positioned at the same height, thus forming a control volume (see Fig. 4) which Eq. (3) was applied for. The tips were aligned with the bottom of the air gap in order to minimize $\delta\dot{q}$. The surface temperature of the calorimeter (T_{surf}) was measured with a K-Type thermocouple soldered in the center of the sensing element with the tip at the surface plane (7, Fig. 4).

The surface temperature of the target varied only about 50 °C Xwhen changing the mass flow of the calorimetric water. This was not enough to study the heat transfer behaviour at different surface temperatures (temperatures of interest were between 200 and 650 °C). Therefore, two calorimeters were designed: the first calorimeter (CU calorimeter) was completely manufactured of CuETP copper (both, the sensing element and the guard). Because of its high thermal conductivity ($388\ \frac{W}{m\,K}$) the surface temperature of the sensing element (T_{surf}) did not exceed 200 °C. The second calorimeter (PSI calorimeter) was also made of CuETP copper, but a 2 mm thick layer made of stainless steel (1.4435, thermal conductivity $15\ \frac{W}{m\,K}$) was soldered 1.5 mm beneath the sensing surface (shown in Fig. 4). It reduced the overall thermal conductivity of the sensing element and thus increased the temperature of its surface up to a maximum of 605 °C. A thin part of the sensing element (1.5 mm) exposed to the flame jet was still made of copper to ensure a uniform surface temperature profile, as in the case of CU calorimeter. A disadvantage of the copper layer exposed to the jet is its oxidation at high temperatures. Therefore, all experiments were performed after the oxidation film had been formed in order to keep physical properties of the calorimeter unchanged. The guard was manufactured from stainless steel in order to keep the construction simple.

The heat transfer coefficient was calculated as in Eq. (5):

$$h = \frac{\dot{q}_{sen}}{(T_f - T_{surf})} \tag{5}$$

where the difference $T_f - T_{surf}$ presents the driving force for the heat transfer from the impinging flame jet to the rock surface. T_{surf} is the temperature of the sensing surface. T_f is the temperature of the flame jet at the exit nozzle (T_{nozzle}) as defined by Baukal et al. [19]. Its measurement was described in chapter 2.1.

2.3. Experimental procedure

During the experiment, the measuring procedure was the following:

- The mass flows of the guard and the calorimetric water were chosen and the chiller was adjusted to reach a desired inlet temperature. Afterwards, the flame was ignited.
- After reaching the desired firing rate, the calorimeter was positioned at the desired dimensionless distance (L) with a linear actuator (L = 5–14). As soon as steady state had been reached the measurement was performed.
- The surface temperature of the sensing element as well as the inlet and the outlet temperature of the calorimetric and guard water were recorded for 30 s.

3. Uncertainty

Devices that are included in the heat transfer study are listed in Table 3 with their measuring uncertainties.

The final uncertainty in measured heat flux ($\delta\dot{q}_{sen}$) and heat transfer coefficient (δh) was calculated using the method of Kline et al. [23]. It is in detail presented in Appendix A. K-type thermocouples, which measured the temperature of the calorimetric water, were calibrated in order to reduce their measuring uncertainty because they had the largest influence on the final uncertainty in the heat flux. The calibration was performed with a FLUKE 9142 Metrology Well and the obtained data of the thermocouples and reference temperatures were fitted with a least squares linear regression method. Any variations in the methane and air flow could also influence the heat transfer study. The uncertainties of the methane and air flow rate measurement were ± 3.70% and ± 0.61%, respectively. The final uncertainty of the heat flux was in the best case 7% and in the worst 45% as shown in Fig. 5. The highest uncertainties were obtained at the lowest heat fluxes because the temperature difference between the inlet and outlet cooling water ($T_{outlet} - T_{inlet}$ in Eq. (3)) was only 1 °C. However, process conditions with heat fluxes higher than $4\ \frac{MW}{m^2}$ are required to perform flame jet drilling. In that range, the uncertainty in obtained heat fluxes was less than 20%. The uncertainty range of the heat transfer coefficient is presented with the results (chapter 4.2) being between ± 8% and ± 45%, but at the process conditions relevant for flame jet drilling the uncertainty was also less than 20%.

4. Results and discussion

The results are divided into four subchapters. The first chapter

Table 3
Uncertainty of measuring devices.

Measured quantity	Symbol [δU_d]	Instrument	Uncertainty
Mass flow of calorimetric water	$\delta\dot{m}$	Flow meter	0.2% [a]
Inlet water temperature	$\delta T_{inlet,d}$	Calibrated K-type thermocouple	0.32 °C
Outlet water temperature	$\delta T_{outlet,d}$	Calibrated K-type thermocouple	0.32 °C
Surface temperature	$\delta T_{surf,d}$	K-type thermocouple	0.075% a
Flame temperature	$\delta T_{f,d}$	S and K-type thermocouple	0.25% and 0.075% a

[a] of measured value

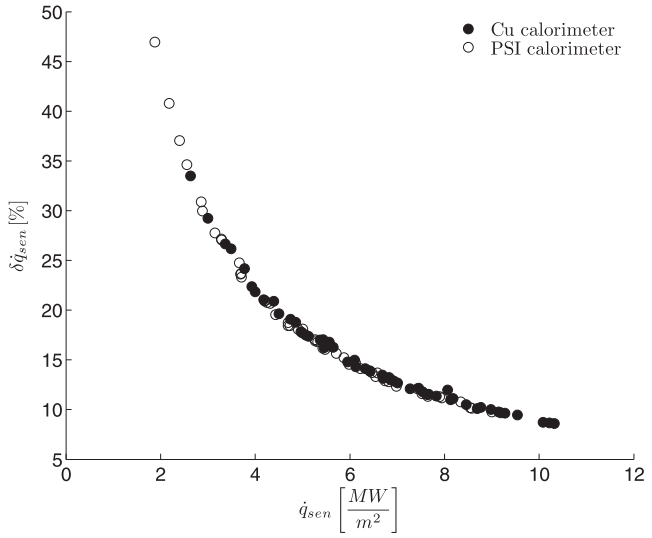


Fig. 5. Uncertainty in heat flux measurement with both calorimeters. PSI calorimeter measured less heat flux than CU calorimeter at same operating conditions, because of its larger thermal resistance.

contains properties of the flame jet and the influence of the confinement on the jet behaviour. Then, in the second chapter, the performance of both calorimeters is presented. The third and fourth chapter describe the influence of the process conditions (firing rate, nozzle diameter and dimensionless distance) on the heat flux transferred between the confined impinging jet and the calorimeter.

4.1. Description of applied flame jet being confined and unconfined

The flame jet was both subsonic (at 23 and 28 kW) and underexpanded supersonic (at 33 and 38 kW) when the smaller nozzle ($D_1 = 7.9$ mm) was installed. The underexpansion of the exhaust gases was clearly registered with an infrared camera as presented in Fig. 6.

The flame jet had a shock cell structure at 33 and 38 kW. The transition to underexpanded jet occurred at firing rate 31 kW when the pressure in the chamber (p_{ch}) reached 2.25 bar. At this point, instead of having a smooth pressure drop after the nozzle exit, the pressure starts oscillating along the jet axis. This periodical expansion and compression happens until the pressure in the exhaust gases becomes equal to the ambient pressure. The further the firing rate was increased, the more the shocks became visible. The camera registers more dense regions (light spots) of the exhaust gases that are related to the pressure oscillations present in the underexpanded jet [24].

Heat transfer from an underexpanded jet is not described much in the literature. Yu et al. [25] were investigating the heat transfer from an underexpanded air jet impinging on a flat surface. It was noticed that the heat flux oscillates with the distance because of the pressure changes in the jet. Those oscillations were also present in the case of applied flame jet in this work when impinging the flat calorimetric surface without the confinement. As shown in Fig. 7, the heat flux measured from the unconfined underexpanded impinging jet was fluctuating with dimensionless distance ($\dot{Q}_f = 38$ kW and $L < 9$). For the firing rate 23 kW the jet was subsonic without shocked structure (see Fig. 6), thus the fluctuations did not exist. At larger distances the shocks were not present, thus no oscillations in the heat flux are visible. A different phenomenon was registered with the confined jet: the confinement seemed to prevent the entrainment of ambient air such that no strong expansion and compression of the exhaust gases could occur. Therefore, no heat flux fluctuations were visible (see Fig. 7). When comparing the confined and unconfined heat flux results, the confined flame jet transferred more heat to the impinged surface. This could be due to reduced entrainment of ambient air in the flame jet having higher momentum of the impinging jet and higher fluid temperature that are beneficial for the heat transfer.

4.2. Influence of calorimeter design on heat transfer from confined flame jet

The influence of target surface temperature on the heat transfer was analyzed using results from the heat transfer study of the confined impinging flame jet, because it is more relevant for the case when drilling a rock sample. The PSI calorimeter had higher thermal resistance than the CU calorimeter, which influenced the surface temperatures of the sensing element. They were between 250 and 600 °C and between 70 and 200 °C for PSI and CU calorimeter, respectively (see Fig. 8). As a consequence, more heat was transferred to the calorimetric water through the CU calorimeter (up to 10.5 $\frac{MW}{m^2}$) than through the PSI calorimeter (up to 9.0 $\frac{MW}{m^2}$) for the same process conditions (see Fig. 9).

The calculated heat transfer coefficient is presented in Fig. 10. For most operating conditions there was no apparent influence of the surface temperature on the heat transfer coefficient. At dimensionless distances smaller than 8 and firing rate of 38 kW, it seems that the calculated heat transfer coefficient was smaller for the CU calorimeter than for the PSI calorimeter. At those cases the amount of transferred heat was very high, thus it could be that one part of the heat transferred through the sensing surface of CU calorimeter was lost: either it had not entered the cooling water in the control volume, or it might have been transferred to the guard ($\delta\dot{q} < 0$). Both could occur because the CU calorimeter had larger overall thermal conductivity than the PSI calorimeter.

4.3. Influence of process parameters on heat flux transferred from confined flame jet

The influence of firing rate, dimensionless distance and nozzle diameters on the heat flux transferred from the impinging flame jet is discussed in this chapter. The presented results were obtained with the PSI calorimeter that was impinged by the confined flame jet.

4.3.1. Influence of firing rate

From Fig. 11 it can be seen that by increasing the firing rate the amount of transferred heat to the calorimeter was increasing. The maximum heat flux was around 9 $\frac{MW}{m^2}$, which is a bit higher than the maximum heat flux obtained by Rauenzahn [8] (7.4 $\frac{MW}{m^2}$). By increasing the firing rate more reactants are combusted, the temperature at the nozzle and the mass flow of the exhaust gases increased, thus, more heat was transferred to the sensor. Moving the calorimeter further away from the exit nozzle, the heat flux decreased because more ambient air was entrained into the jet influencing the temperature and also the jet momentum before it impinged the sensor surface. Additionally, the entrainment was more intense at higher firing rate due to shock structure of the jet, which resulted in higher heat flux gradient with

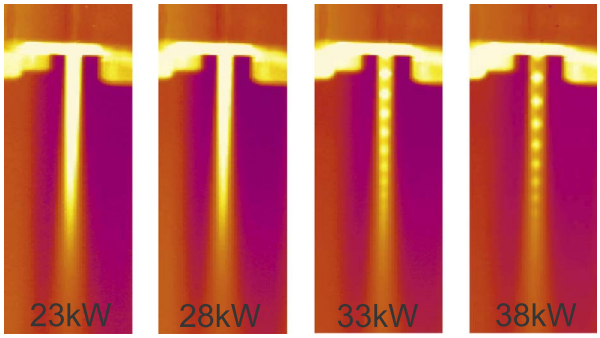


Fig. 6. Free jet observed with infrared camera at different firing rates with the smaller exit nozzle ($D_1 = 7.9$ mm). Underexpanded supersonic jet flow (shocked structure) was present at 33 and 38 kW.

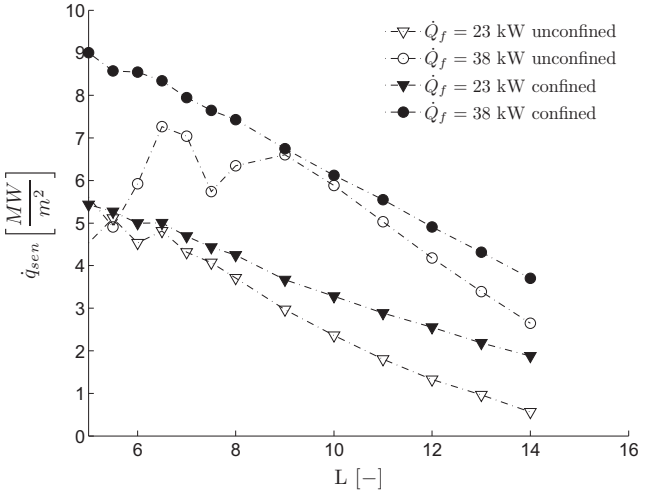


Fig. 7. Heat flux obtained with the PSI calorimeter with and without confinement. The nozzle diameter was D_1 = 7.9 mm at different firing rats.

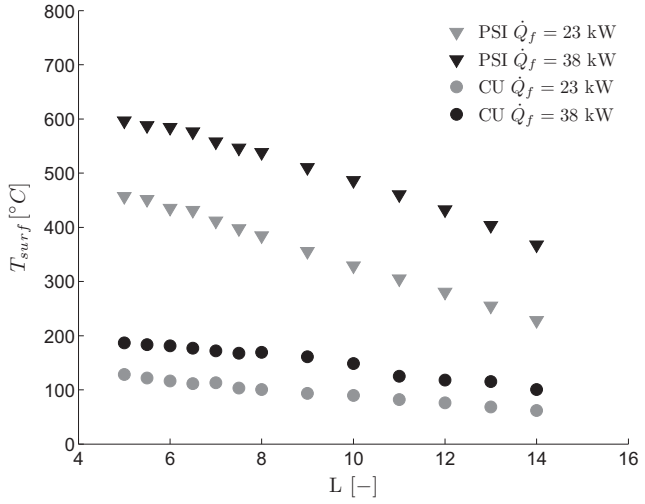


Fig. 8. Surface temperatures of the PSI and CU calorimeter at different dimensionless distances and at different firing rates. Nozzle diameter was D_1 = 7.9 mm and the flame jet was confined.

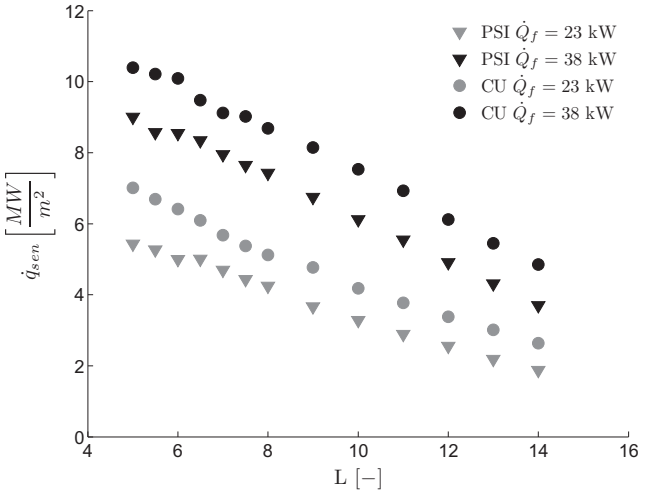


Fig. 9. Heat flux measured with the PSI and CU calorimeter at different dimensionless distances and at different firing rates. Nozzle diameter was D_1 = 7.9 mm and the flame jet was confined.

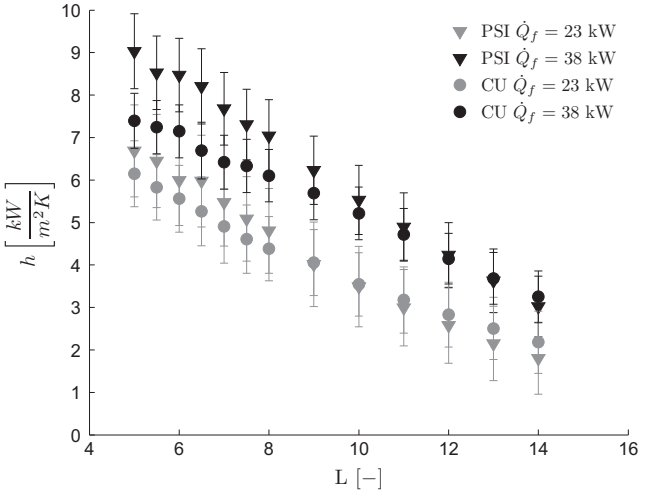


Fig. 10. Heat transfer coefficient calculated from the measurement performed with the PSI and CU calorimeter at different dimensionless distances and different firing rates. The nozzle diameter was D_1 = 7.9 mm and the flame jet was confined.

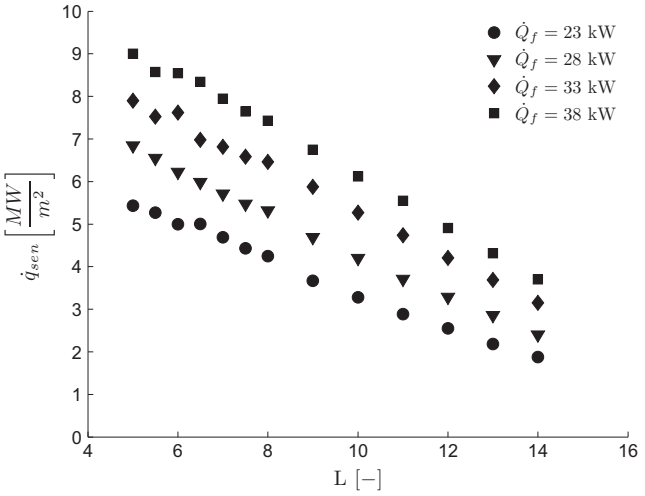


Fig. 11. Heat flux measured with the PSI calorimeter at different dimensionless distances and at different firing rates. Nozzle diameter was D_1 = 7.9 mm and the flame jet was confined.

dimensionless distance.

The heat flux increased linearly with the firing rate having a higher gradient at smaller distances (see Fig. 12). The change in gradient can again be attributed to the air entrainment.

Even though the flame jet was confined, the obtained results were in agreement with the heat transfer study of free impinging flame jets presented in the work of Baukal et al. [16]. The heat flux increased as the dimensionless distance decreased. Also, keeping the distance and the oxidizer composition constant the heat flux increased with the firing rate.

4.3.2. *Influence of nozzle diameter*

For the same firing rate lower heat fluxes were recorded with the exit nozzle of a larger diameter. The reason for the reduced amount of transferred heat is the reduced gas velocity at the nozzle, thus, the convection to the calorimeter is less efficient. This is shown in Fig. 13 by comparing the heat fluxes obtained at 38 kW with both nozzles of diameter D_1 and D_2 ($D_1 < D_2$). At $L = 5$, where the flame jet was completely confined, the heat flux with D_1 was 2.4 times larger than the heat flux with D_2 (9.0 $\frac{MW}{m^2}$ compared to 3.8 $\frac{MW}{m^2}$). The nozzle diameters do

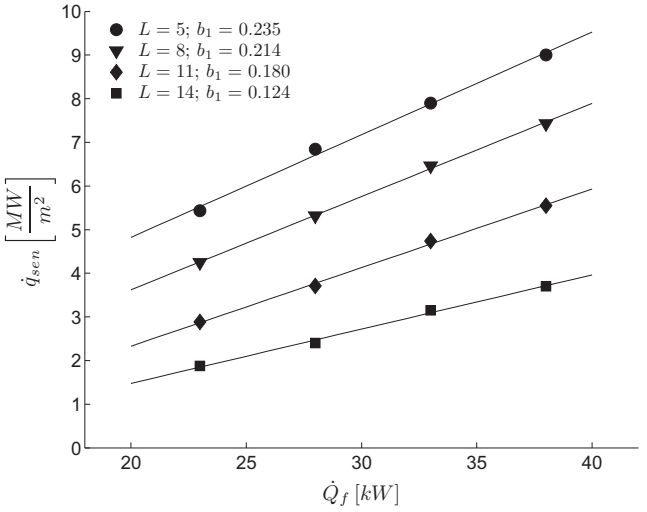


Fig. 12. Heat flux measured with the PSI calorimeter at different firing rates keeping the dimensionless distance constant. Data were fitted with a linear dependancy between the firing rate ($\dot{Q}_f$) and the heat flux ($\dot{q} = b_0 + b_1 \cdot \dot{Q}_f$). Nozzle diameter was D_1 = 7.9 mm and the flame jet was confined.

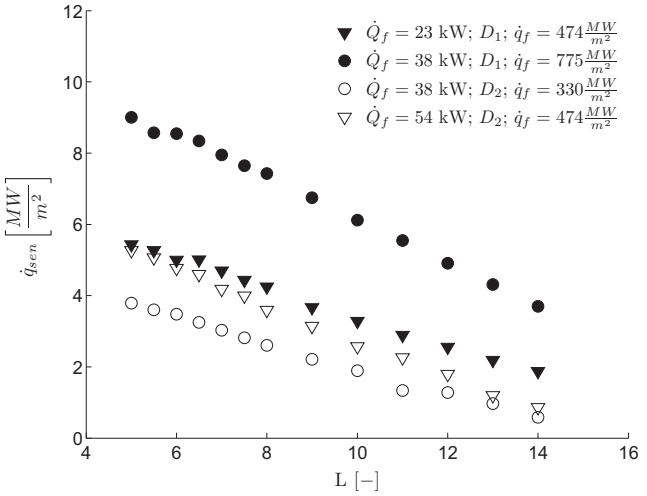


Fig. 13. Heat flux measured with the PSI calorimeter at different dimensionless distances and at different firing rates. Nozzle diameters were D_1 = 7.9 mm, D_2 = 12.1 mm and the impinging jet was confined. Relative firing rate ($\dot{q}_f$) was calculated as in Eq. (6).

not differ by 2.3 times but the nozzle areas do (115 mm^2 compared to 49 mm^2), which the recorded heat flux is directly related to. Fig. 13 also denotes the relative firing rate $\dot{q}_f$, which is the ratio between the firing rate $\dot{Q}_f$ and nozzle area A_{nozzle} as calculated in Eq. (6)

$$\dot{q}_f = \frac{\dot{Q}_f}{A_{nozzle}} \tag{6}$$

With a firing rate of 23 kW and the small nozzle diameter D_1 the relative firing rate was the same as with the larger nozzle (D_2) at 54 kW ($\dot{q}_f = 474\ \frac{\mathrm{MW}}{\mathrm{m}^2}$). For the same relative firing rate the recorded heat fluxes were equal at smaller dimensionless distances, where the flame jets were completely confined.

4.3.3. Comparison of all influencing parameters

The maximum heat flux was measured at $\dot{Q}_f(\dot{q}_{max})$ = 38 kW, $D(\dot{q}_{max})$ = 7.9 mm ($\dot{Q}_f = 775\frac{\mathrm{MW}}{\mathrm{m}^2}$) and $L(\dot{q}_{max})$ = 5. It was noticed that the influence of change in distance on heat flux is smaller compared to the influence of a change in relative firing rate when being close to the exit nozzle. The influence of a change in distance becomes more significant when positioned further away from the exit nozzle. As an example, in order to halve the recorded heat flux when the target is placed close to the nozzle (L = 5), the relative firing rate has to be decreased also by half. But in order to halve the heat flux by changing the distance, L has to be increased up to 14, which is almost 3 times larger. When the target is placed further away from the nozzle (L = 10) the heat flux can still be halved by halving the relative firing rate, but a change in the distance from 10 to 14 has the same effect. This change in distance is smaller than when the target is close to the exit nozzle. Therefore, the further away the target is located from the exit nozzle the more the changes in dimensionless distance gain influence on the heat flux.

5. Conclusion

The heat transfer study performed with both subsonic and supersonic underexpanded impinging flame jet can be concluded with the following:

- The underexpanded free flame jet impinging on the calorimeter generated a heat flux that fluctuated with dimensionless distance. This was a consequence of the generated shocks in the flow of the exhaust gases. By confining the flame jet no oscillations have been observed due to limited jet expansion (shock absence).
- Two calorimeters have been constructed for the reported heat transfer study: PSI and CU calorimeter. The PSI calorimeter was more heat resistant, thus, less heat was transferred to the calorimetric water. This resulted in higher surface temperatures of the sensing element.
- The results showed that at different surface temperatures of the target (being up to 600 °C) did not remarkably influence the heat transfer from the same impinging flame jet.
- For future calorimeter designs, it is suggested to increase the length of the observed control volume in the CU calorimeter, or to increase the sensing surface in order to reduce the possible heat losses.
- The process parameters that are related to the impinging flame (the firing rate and the nozzle diameter) had a larger influence on the heat flux transferred to the target placed very close to the exit nozzle, than the dimensionless distance. If the target was further away from the nozzle, all process parameters had an equal impact on the measured heat flux.

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Appendix A. Error calculation

Eqs. (7) and (8) present the error calculation of heat flux and heat transfer coefficient, respectively. Because the measured data was recorded for a certain period of time, the standard deviation σU of each measurement U ($U \in \{\dot{m}, T_{inlet}, T_{outlet}, T_{surf}, T_f\}$) could be determined and included in the uncertainty calculation as presented in Eq. (9).

$$\delta\dot{q} = \sqrt{\left(\frac{C_p(T_{outlet}-T_{inlet})}{A}\delta\dot{m}\right)^2 + \left(\frac{\dot{m}C_p}{A}\delta T_{outlet}\right)^2 + \left(-\frac{\dot{m}C_p}{A}\delta T_{inlet}\right)^2} \tag{7}$$

$$\delta h = \sqrt{\left(\frac{C_p(T_{outlet}-T_{inlet})}{A(T_f-T_{surf})}\delta\dot{m}\right)^2 + \left(\frac{\dot{m}C_p}{A(T_f-T_{surf})}\delta T_{outlet}\right)^2 + \left(-\frac{\dot{m}C_p}{A(T_f-T_{surf})}\delta T_{inlet}\right)^2 + \left(-\frac{\dot{m}C_p(T_{outlet}-T_{inlet})}{A(T_f-T_{surf})^2}\delta T_f\right)^2 + \left(\frac{\dot{m}C_p(T_{outlet}-T_{inlet})}{A(T_f-T_{surf})^2}\delta T_{surf}\right)^2} \tag{8}$$

$$\delta U = \sqrt{(\sigma U)^2 + (\delta U_d)^2} \tag{9}$$

Uncertainty of the measuring devices was presented with δU_d and it is listed in Table 3.

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